Name: ______________________

# RATIO,
# (RATIO AND PROPORTION PROBLEMS)

Date: _______________

**LO: I can solve multi-step problems involving ratio and proportion.**

## Ratio and proportion problems

**Ratio** compares parts to parts. **Proportion** compares a part to the **whole**. Both are used to solve sharing and scaling problems.

**Strategy:** Identify what you know, find one part or unit, then scale to find what is needed.

## Quick examples

| | | |
|---|---|---|
| ● | Ratio 2:3 proportions are $\frac{2}{5}$ and $\frac{3}{5}$ | convert ratio to fractions |
| ● | If 3 parts = 45, then 1 part = 15 | unitary method |
| ● | Scale factor = new ÷ original | for enlargement |
| ● | Ratio 2:3 means $\frac{2}{3}$ of the total | it is $\frac{2}{5}$ of the total |
| ● | Ratio problems always have whole-number answers | not always |

## Key idea

Convert between ratio and proportion: ratio a:b means fractions a/(a+b) and b/(a+b) of the total.

**Ratio a:b fractions a/(a+b) and b/(a+b)**

**Sum the parts to find the whole**

## MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

### Example 1 – ratio to proportion

In a class, boys to girls = 3:5. What fraction are boys?

Total parts = 3 + 5 = 8

Boys = $\frac{3}{8}$ of the class

If 32 students, boys = $\frac{3}{8}$ × 32 = 12

**Boys are $\frac{3}{8}$ of the class**

The ratio 3:5 means 8 parts in total.

### Example 2 – given one part

Purple paint is mixed red:blue = 2:5. You have 400 ml of blue. How much red?

Blue = 5 parts = 400 ml

1 part = 400 ÷ 5 = 80 ml

Red = 2 parts = 2 × 80 = 160 ml

**160 ml of red**

Use the known quantity to find one part.

### Example 3 – difference given

Two numbers are in ratio 3:8. Their difference is 30. Find the numbers.

Difference in parts = 8 - 3 = 5 parts

5 parts = 30, so 1 part = 6

Numbers: 3 × 6 = 18 and 8 × 6 = 48

**18 and 48**

The difference between the ratio parts helps find one part.

### Example 4 – multi-step

A drink is made from concentrate and water in ratio 1:4. 750 ml of drink is needed.

Total parts = 1 + 4 = 5

One part = 750 ÷ 5 = 150 ml

Concentrate = 150 ml, water = 600 ml

**150 ml concentrate, 600 ml water**

The total drink amount is split into ratio parts.

## YOUR TURN – PRACTICE (deliberate practice)

### 1. Ratio to fractions

Write each ratio as fractions of the whole.

**a) 1:3**

**b) 2:5**

**c) 4:1**

**d) 3:3:4**

### 2. Finding amounts from one part

Given one part of the ratio, find the other.

**a) Ratio 2:7. Smaller = 14. Find larger.**

**b) Ratio 3:4. Larger = 60. Find smaller.**

**c) Ratio 1:5. Smaller = 8. Find total.**

**d) Ratio 5:2. Total = 63. Find each.**

### 3. Difference problems

The difference between two parts is given. Find the amounts.

**a) Ratio 2:5, difference = 21**

**b) Ratio 1:4, difference = 36**

**c) Ratio 3:7, difference = 20**

**d) Ratio 5:9, difference = 32**

### 4. In context

Solve each problem.

**a) Mortar: cement to sand 1:6. Need 3.5 kg total. Cement?**

**b) Boys:girls 5:4. 36 students total. How many boys?**

**c) Money split 3:2. Larger share is £45. Total?**

**d) Recipe: sugar to flour 2:3. Use 600 g flour. Sugar?**

## COMMON MISCONCEPTIONS – SPOT THE MISTAKE

**Each statement shows a student's answer. Tick ( ) the ones that are correct.** If it is wrong, write the correct answer.

**a) Ratio 3:5 means 3 parts out of 8 total** ☐

Correct answer: __________

Reason: __________________

**b) Ratio 3:5 means 3 out of 5** ☐

Correct answer: __________

Reason: __________________

**c) If the difference is given, subtract ratio numbers to find one part** ☐

Correct answer: __________

Reason:

**d) Ratio 2:3 and proportion ⅔ mean the same thing** ☐

Correct answer: __________

Reason: __________________

**e) You can compare ratios by writing equivalent ratios** ☐

Correct answer: __________

Reason: __________________

## 6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) In a school, the ratio of boys to girls is **5:4**. If there are **24 more boys** than girls, how many students are in the school?

**Expression:** ______________________________

**Simplified:** ______________________________

b) A recipe uses flour, sugar, and butter in the ratio **5:3:2**. You have **800 g of flour**, **500 g of sugar**, and **400 g of butter**. What is the maximum amount of the recipe you can make? Which ingredient limits you?

**Expression:** ______________________________

**Simplified:** ______________________________

### TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

### CHECK YOUR WORK

**Have I shown all my working clearly?**

**Did I check my answer using a different method?**

**Does my answer look reasonable?**

WELL DONE!